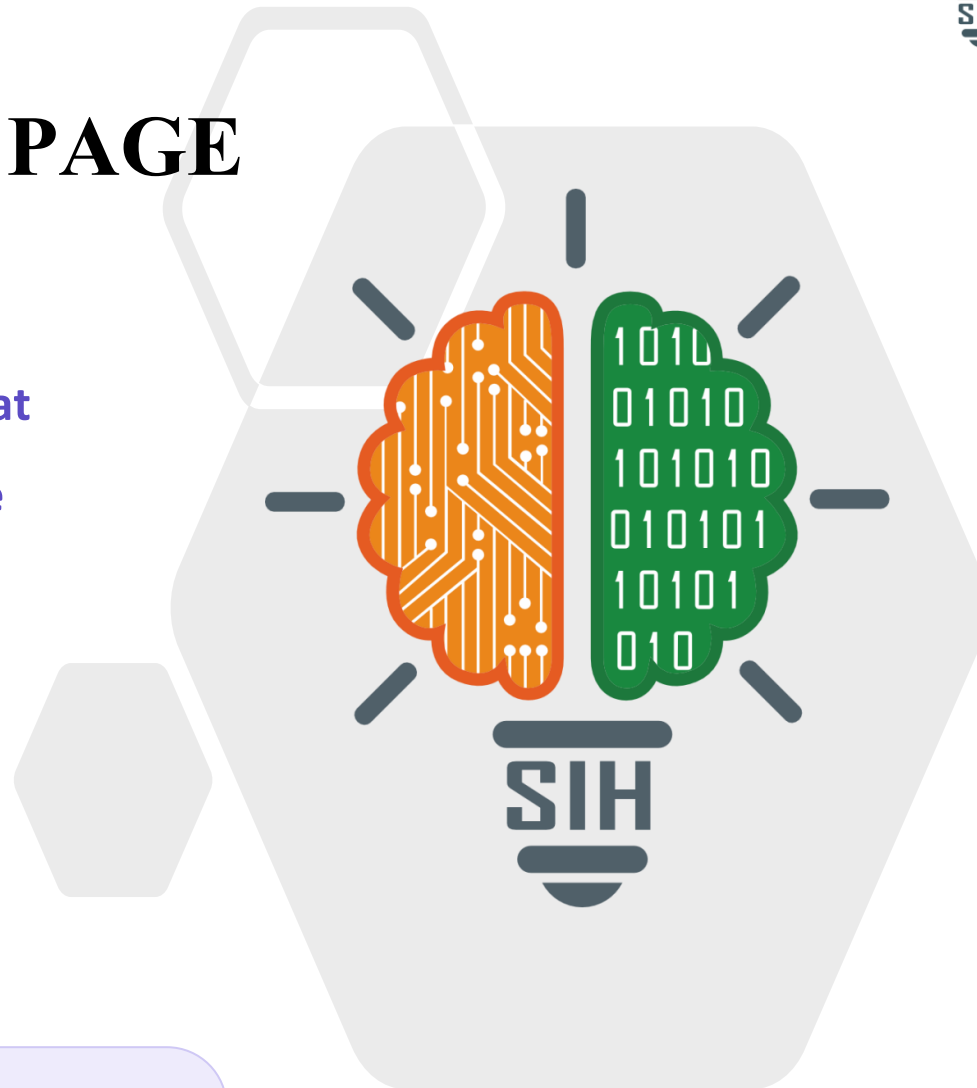


TITLE PAGE

- Problem Statement ID – **26106**
- Problem Statement Title- **AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID- **142647**
- Team Name- Red Eye



Our solution: Security-Copilot · Detect → Trace → Attribute → Report

 The Problem

\$3.04 B lost to Business Email
Compromise in 2025
(FBI IC3)

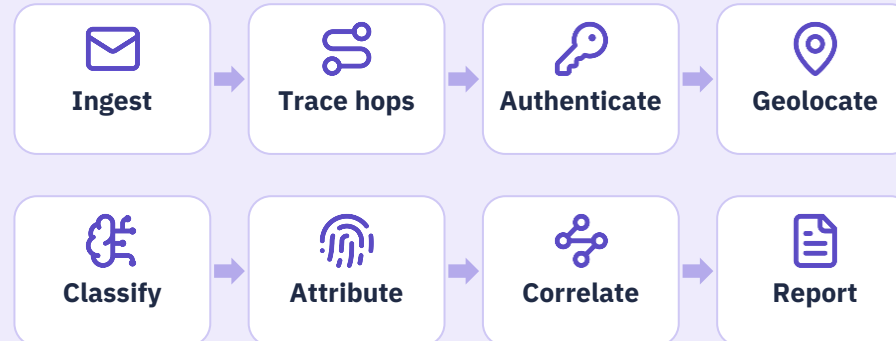
₹22,495 Cr lost by Indians to cyber
fraud in 2025 (MHA /
I4C)

The gap

- Filters only block or deliver — no forensic trace of **where** it came from
- Headers, SPF/DKIM/DMARC, IP & WHOIS checked by hand, in 5+ separate tools
- AI-written lures, lookalike domains & relay chains slip past static rules
- Related frauds are never linked into **campaigns**

 Proposed Solution

Drop in a raw email (.eml), a URL or pasted text. An **AI agent investigates it like an analyst** — traces its route, verifies the sender, geolocates the origin and checks every link & attachment — then returns a **classified, attributed, explainable verdict** with a forensic report.



Answers in the five classes the PS asks for:

legitimate **suspicious** **impersonated** **phishing** **fraud-related**
+ 0–100 risk score · attribution · origin map · campaign ID

 Why it's different

-  **Investigates, not just filters**
An agent chooses its own tools & depth
-  **Explainable risk score**
Every point carries a written reason
-  **Attribution, not guesswork**
Spoofed · compromised · anonymised · direct actor
-  **Remembers past cases**
SecureBERT memory clusters campaigns
-  **Private & lawful by design**
PII masked, evidence kept, report—never hack back

 Not just **“is this phishing?”** — but **where did it come from, who sent it, and what else is it connected to?**

1 ENTRY POINTS



Chrome MV3 extension
every page visit · auto-check of
each Gmail message opened



Gmail Add-on
Apps Script auto-scanner labels
the inbox · sidebar report card



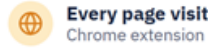
Analyst dashboard
.eml upload · case pages ·
campaigns · PDF report



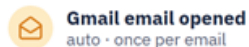
Python CLI
URL & raw-email checks
from the terminal

2 FAST TRIAGE

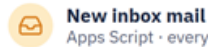
0 false alerts in live Gmail tests



Every page visit
Chrome extension



Gmail email opened
auto · once per email



New inbox mail
Apps Script · every min

LINK CHECK URLs & email links

ONNX URL model
local ML phishing score

VirusTotal corroboration
cached · can overrule the model

link verdict → context for Jev

TEXT VERDICT email body

Jev PRIMARY CLASSIFIER
decides the text verdict

BERT phishing model
local backup if Jev is offline

combined verdict



Banner · Gmail label

Full scan on demand

full scans: dashboard · CLI · extension · Gmail "Check Report"

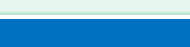
4 EVIDENCE, PRIVACY & CASE MEMORY



SQLite case history
raw headers kept as
evidence



SecureBERT embeddings
case memory



PII masking
Microsoft Presidio



NetworkX entity graph
sender · domain · IP



DBSCAN campaigns
related frauds clustered



90-day retention
auto-delete

3 FULL INVESTIGATION

LangGraph agent · live progress over SSE

FastAPI
SSE live progress

LangGraph
stateful workflow

Link cache
blocklist · 24 h

Raw email?
headers present?

YES → deterministic forensics



Header & auth forensics

- Received hops → hop-0 origin
- SPF · DKIM · DMARC + alignment
- Sender-domain age · WHOIS
- Origin IP · TOR / VPN / hosting



Attachment & PDF sandbox

- Hash · real type · macros · ClamAV
- PDF text → phishing classifier
- PDF links → ML + VirusTotal
- Embedded JS · auto-launch · files
never opened, rendered or executed

no headers → straight to the agent



LLM AGENT via OpenRouter
Reasons over all evidence, picks the next tool, loops until confident.
Outage or limit → fails safe to "suspicious", never "legitimate".

Investigation tools called by the agent

Sandboxed browser
headless Chromium

Brand search
DuckDuckGo

WHOIS + VirusTotal
domain reputation

ONNX + BERT
URL & text models

IP geolocation
GeoLite2 · TOR / VPN

Entity correlation
NetworkX graph

SecureBERT case recall
similar past investigations

5 ANALYST-FACING RESULT



Explainable verdict

legitimate suspicious
impersonated
phishing
fraud-related

0–100 risk score, a written
reason for every point, and
attribution



Dashboard
case page · map · graph

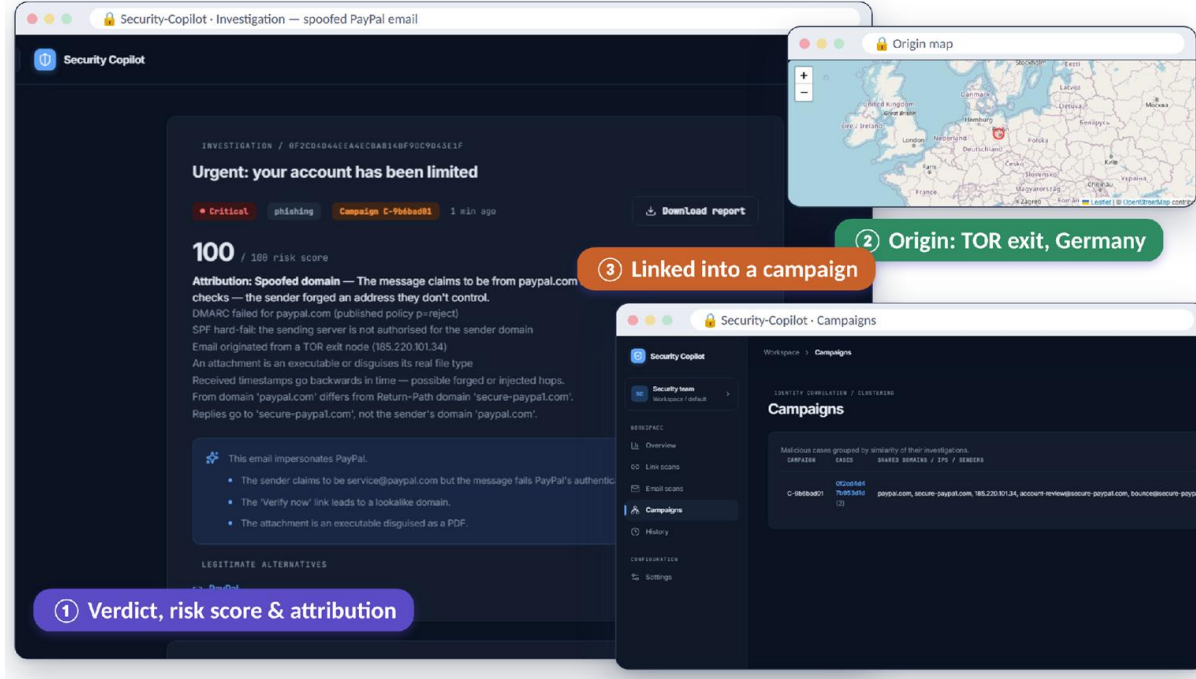


Gmail
colored labels + sidebar card

Extension
warn · report · block



Dangerous site
Report & block

BUILT AND RUNNING — the Security-Copilot web app on a real fraud case

0

false alerts in
live Gmail tests

6 / 6

PS modules
delivered

12

AI & forensic
tools

26

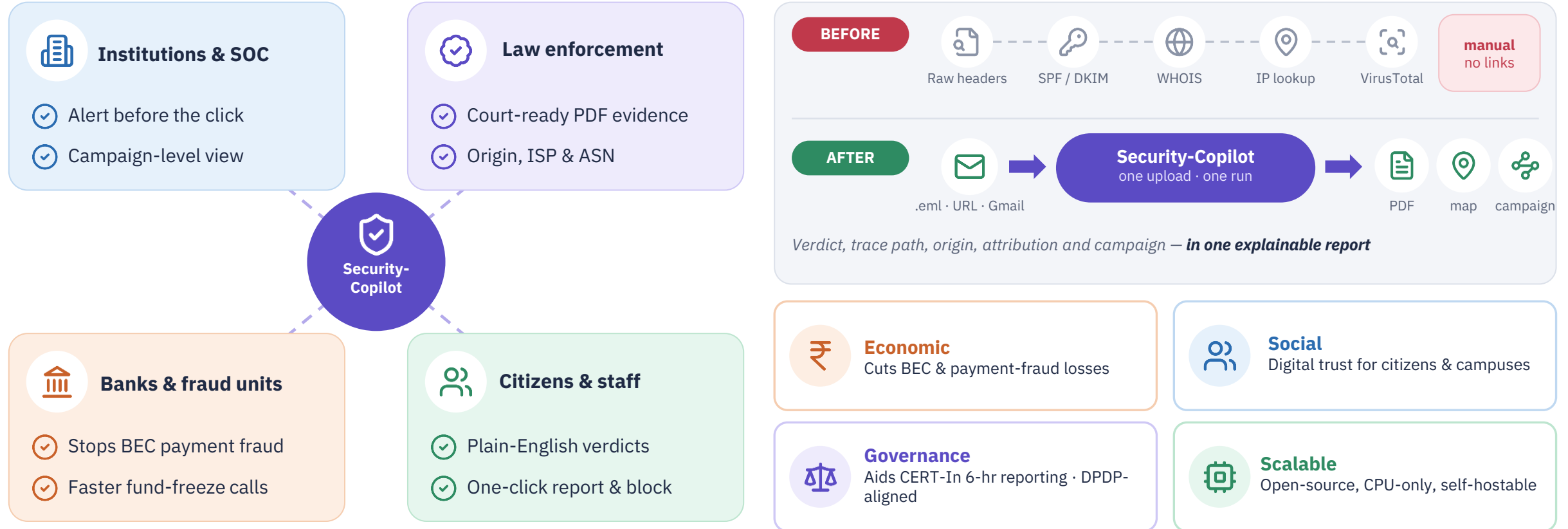
automated tests
passing

₹0

paid threat-intel
APIs required**KEY CHALLENGE** → **HOW WE HANDLE IT****TOR / VPN hides origin**Flag as anonymised; origin shown as
“probable”**Flawless AI-written
lures**Jev judges intent, not grammar;
headers & links checked apart**LLM error or outage**Rule-based score; fails safe to
“suspicious”**Weaponized
attachments**Sandboxed read of text, links &
scripts — never opened or run**Sensitive personal data**

PII masked · 90-day auto-delete

**API cost & rate limits**Offline GeoIP + cache; paid APIs
optional**DKIM keys / DNS
change**Receiver's verdict kept beside live
re-check



ROADMAP





1. Email authentication standards

RFC 7208 — SPF

datatracker.ietf.org/doc/html/rfc7208

RFC 6376 — DKIM Signatures

datatracker.ietf.org/doc/html/rfc6376

RFC 7489 — DMARC

datatracker.ietf.org/doc/html/rfc7489

RFC 5322 — Message format & Received trace

datatracker.ietf.org/doc/html/rfc5322



2. AI / ML models & methods

Jev (typesafe) — primary phishing classifier

openrouter.ai/api/alpha/decisions

LangGraph — stateful agent orchestration

github.com/langchain-ai/langgraph

BERT phishing classifier (ealvaradob)

huggingface.co/ealvaradob/bert-finetuned-phishing

ONNX URL model (pirocheto)

huggingface.co/pirocheto/phishing-url-detection

SecureBERT, Aghaei et al. 2022

arxiv.org/abs/2204.02685



3. Threat intel & geolocation

MaxMind GeoLite2 City / ASN

dev.maxmind.com/geoip/geolite2-free-geolocation-data

Tor Project bulk exit list

check.torproject.org/torbulkexitlist

X4BNet VPN & datacenter ranges

github.com/X4BNet/lists_vpn

AbuseIPDB · VirusTotal APIs

abuseipdb.com · virustotal.com



4. Forensic & privacy tooling

checkdmrc · dkimpy (SPF/DMARC/DKIM)

pypi.org/project/checkdmrc

oletools — Office macro analysis

github.com/decalage2/oletools

pypdf · ClamAV — PDF sandbox & signature scan

pypi.org/project/pypdf · clamav.net

Microsoft Presidio — PII masking

microsoft.github.io/presidio

DBSCAN, Ester et al., KDD 1996 · NetworkX

networkx.org



5. Threat landscape data

FBI IC3 Internet Crime Report 2025

ic3.gov/AnnualReport/Reports/2025_IC3Report.pdf

MHA / I4C: ₹22,495 Cr cyber-fraud loss, 2025

theprint.in (Feb 2026)

CERT-In Directions, 28 Apr 2022 (6-hr reporting)

cert-in.org.in

Digital Personal Data Protection Act, 2023

meity.gov.in



6. Our platform

Source code — Security-Copilot (SIH26106)

<https://github.com/code-YK/Security-Copilot-Sih26>

Backend: FastAPI + LangGraph agent

26 automated tests · pytest

Dashboard: Next.js · Extension: Chrome MV3

screens: case page, graph, campaigns

Gmail Add-on: Google Apps Script

developers.google.com/workspace/add-ons

Case study: spoofed PayPal email via TOR

risk 100 · campaign C-9b6bad01